Quality Control Layer 1: Research & Factual Accuracy Audit

Lesson: HTTP Requests and Responses | **Audit Reference:** AUDIT-FACT-001

Overall Audit Status: PASS **Factual Accuracy Score:** 96.0% **Issues Detected:** 4

1. Executive Accuracy Assessment

The lesson is factually accurate regarding HTTP semantics, GET/POST method properties, status code classes, and debugging practices. All core claims align with RFC 9110 and MDN Web Docs. Minor issues include a slight imprecision in the 403 Forbidden description (conflating authentication with authorization), an overstatement about curl -v showing the 'full handshake,' and reliance on a book source available only as metadata/preview. No critical or high-severity inaccuracies were found.

2. Factual Claims Analysis & Corrections (4 items)

ID / Severity	Affected Section & Claim	Audit Finding & Supporting Evidence	Recommended Correction
FACT-001 MEDIUM	**Section:** Section 3: Key Terminology / Section 4: Status Code Categories *"403 Forbidden (authenticated but not allowed)"*	RFC 9110 Section 15.5.3 defines 403 Forbidden as the server understanding the request but refusing to authorize it. It does not require that the client be authenticated; 403 may be returned regardless of authentication state (e.g., insufficient permissions, IP-based blocking, or resource-level policy). Describing it as 'authenticated but not allowed' conflates authentication with authorization and could mislead learners about the distinction between 401 and 403. **Evidence:** RFC 9110 Section 15.5.3 (403 Forbidden); MDN Web Docs: HTTP response status codes	**Fix:** Replace with: '403 Forbidden (server understands the request but refuses to authorize it; authentication may or may not have occurred)'

FACT-002 LOW	**Section:** Section 4: API Debugging Practices *"Use curl with verbose mode: `curl -v` shows the full handshake, headers, and status line."*	`curl -v` (verbose) displays connection negotiation, TLS/SSL handshake details for HTTPS, request and response headers, and the status line. However, it does not show the raw TCP/IP three-way handshake (SYN, SYN-ACK, ACK) at the packet level; that requires lower-level tools such as tcpdump or Wireshark. The phrase 'full handshake' could be misinterpreted as packet-level capture. **Evidence:** curl manual page / curl documentation (--verbose)	**Fix:** Replace with: '`curl -v` shows connection details, TLS handshake information (for HTTPS), request/response headers, and the status line.'
FACT-003 LOW	**Section:** Section 4: Sources / Section 10: Additional Resources *"Sources: [RFC 9110: HTTP Semantics], [MDN Web Docs], [HTTP: The Definitive Guide]."*	The lesson cites 'HTTP: The Definitive Guide' as a source, but the provided research source indicates this book is accessible only as metadata/preview, not full text. Specific technical claims cannot be verified against the full text of this book. While the lesson content aligns with RFC 9110 and MDN, citing an unverifiable full-text source reduces research integrity. **Evidence:** Source [3] metadata: HTTP: The Definitive Guide — Accessible: Metadata/Preview only	**Fix:** Remove 'HTTP: The Definitive Guide' from inline source citations unless specific page-referenced claims are verified from full text; list it as supplementary reading only in Additional Resources.
FACT-004 LOW	**Section:** Section 3: Key Terminology / Section 4: Status Code Categories *"401 Unauthorized (authentication required)"*	RFC 9110 Section 15.5.2 defines 401 Unauthorized as indicating the request has not been applied because it lacks valid authentication credentials for the target resource. 'Authentication required' is broadly acceptable but less precise than 'missing or invalid authentication credentials'; the term 'unauthorized' is historically misleading because the code actually concerns authentication, not authorization. **Evidence:** RFC 9110 Section 15.5.2 (401 Unauthorized)	**Fix:** Replace with: '401 Unauthorized (request lacks valid authentication credentials)'

3. Verification Scope & Limitations

Source [3] (HTTP: The Definitive Guide) is available only as Google Books metadata/preview; specific page-level claims from that book could not be verified against full text. Source [4] (YouTube video) was referenced only by snippet; no transcript or frame-level verification was performed. All technical claims in the lesson were cross-checked against RFC 9110 (Full Text) and MDN Web Docs (Full Text).

4. Conclusion & Certification

The lesson is factually sound and well-aligned with RFC 9110 and MDN Web Docs. Four minor issues were identified, primarily concerning precise terminology for 401/403, a slight overstatement about curl -v, and citation of a preview-only book. No critical inaccuracies, unsupported claims, or contradictory information were detected.